

DiscreteEventSimulation

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400.0 |
Generated: | Engine: DiscreteSim | Sections: 8

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Overview

A discrete-event simulation study: the engine, the calibration and the experiments, with every number carrying its confidence interval.

MODELS
5

THROUGHPUT
0.133 items_per_hour

CYCLE TIME
MEAN
552.5 minutes

UTILISATION
0.863

COMPLETED
477.5

VALIDATION
validated

REEVALUATION
keep

Design of the study

Seed 20260101
Days 45.0
Replications 8
Horizon 4000
Warmup 400.0
Models [:mmc, :transfer_line, :machine_shop, :inventory, :call_center]
Objective cycle_time_mean
Online policy offline_first

The models

Catalogue

Model	Title	Entity	Resource	Answers
mmc	ServicePool	customer	server	ServicePoolWithIdenticalServers
transfer_line	TransferLine	part	station	SerialLineWithBuffersAndFailures
machine_shop	JobShop	job	machine	JobShopWithRoutingsReworkScrapAndBreakdowns
inventory	InventoryPosition	order	stock	ReorderPointPolicyWithLeadTimeAndBackorders
call_center	ContactCenter	call	agent	AgentPoolWithPatienceAndPriority

Resources

Resource	Kind	Servers	Discipline	Utilisation	Availability	Queue length	Wait	Service	Served	Breakdowns
mill	machine	1	priority	0.904	1.0	5.674	65.8	9.771	332	0
drill	machine	1	priority	1.0	1.0	33.6	277.8	8.2	440	0
grinder	machine	1	priority	1.0	1.0	35.1	315.7	9.682	371	0
lathe	machine	1	priority	0.742	1.0	0.793	9.547	8.957	299	0
press	machine	1	priority	0.589	1.0	0.479	6.022	7.399	286	0

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
wait_mean	552.5 none	12.7 none	minutes	2.294
wait_p95	1034 none	103.3 none	minutes	9.987
cycle_time_mean	552.5 none	12.7 none	minutes	2.294
service_mean	— none	— none	minutes	—
completed	477.5 none	9.543 none	count	1.998
throughput	0.133 none	0.00265 none	items_per_hour	1.998
wip_mean	4.155 none	0.061 none	count	1.476
tardiness_mean	449.0 none	11.5 none	minutes	2.564
scrapped	22.9 none	4.057 none	count	17.7
reworked	71.2 none	5.031 none	count	7.061
setup	2.0 none	0 none	count	0
utilisation	0.863 none	0.011 none	ratio	1.285
availability	1.0 none	0 none	ratio	0
max_queue_length	44.1 none	6.49 none	count	14.7
queue_length_mean	75.5 none	0.054 none	count	0.071
span	3599 none	0.388 none	count	0.011
events	7762 none	105.8 none	count	1.363

The engine

A next-event calendar, coroutines that suspend on two channels, and statistics that know whether they are adequate.

The run

Model `machine_shop`

Time unit `minutes`

Seed `3690686459077356`

Events `7631`

Clock `3999`

Processes `618`

Statistics `13`

Streams `5`

Trace rows 6223
Trace dropped 0
Stop reason horizon

Where the processes ended up

READY
0

RUNNING
0

WAITING
4

BLOCKED
77

PASSIVE
0

FINISHED
537

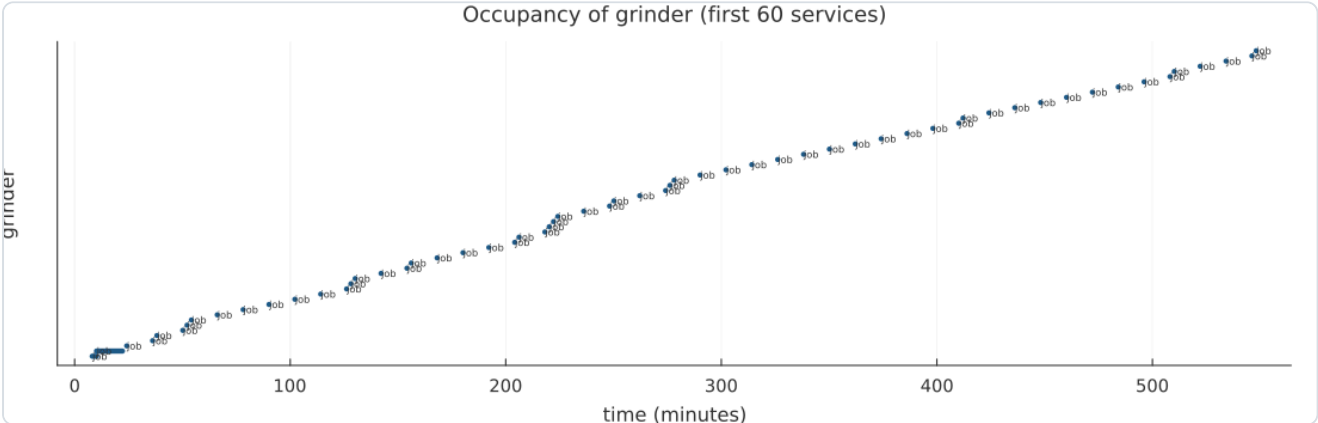
FAILED
0

CANCELLED
0

Little's law, checked on every resource

Little law

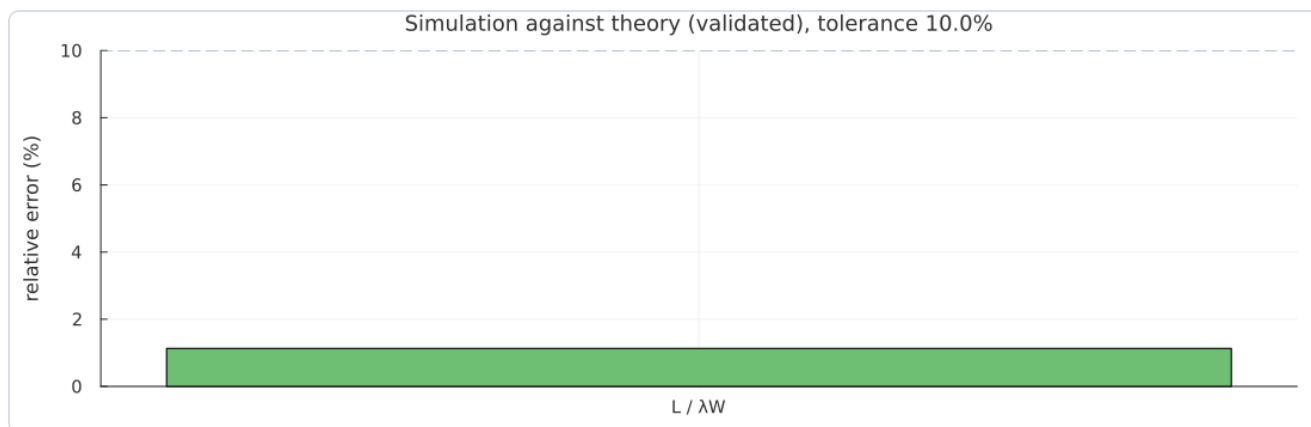
Resource	L	Λ	W	ΛW	Relative error	Holds
mill	6.578	0.092	75.6	6.971	0.06	no
drill	34.6	0.122	286.0	35.0	0.011	yes
grinder	36.1	0.103	325.4	33.5	0.07	no
lathe	1.535	0.083	18.5	1.537	0.00121	yes
press	1.069	0.079	13.4	1.066	0.00208	yes



Occupancy

Against theory

Verdict: validated worst relative error 1.13% against a tolerance of 10.0%.



Validation

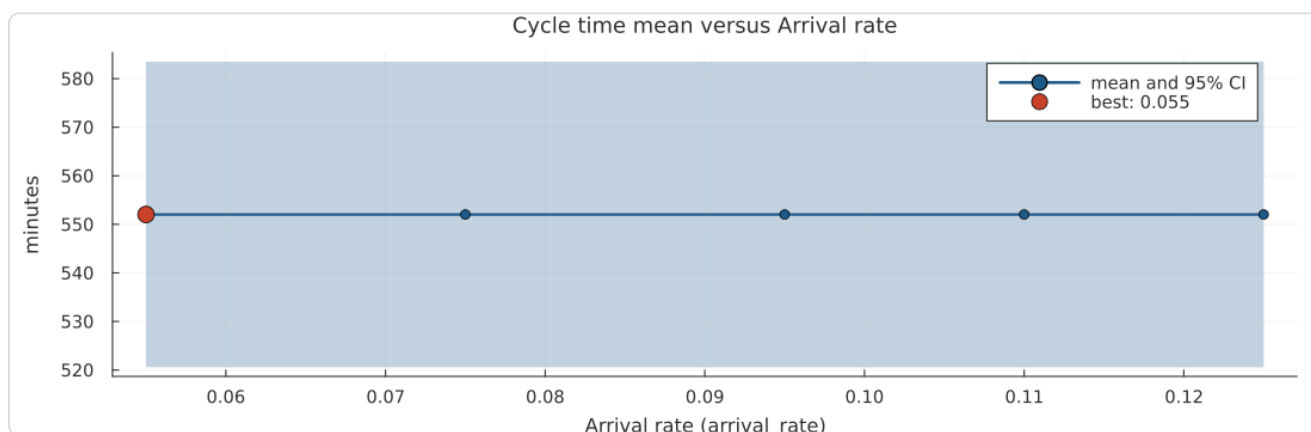
Cross checks

Observed key	Theory key	Observed	Theory	Relative error	Within tolerance
L	λW	34.6	35.0	0.011	yes

Queues and capacity

The first thing to check in a queueing model is that it agrees with Erlang, and the second is what happens when the load moves.

Widening the bottleneck



Sweep

Arrival rate

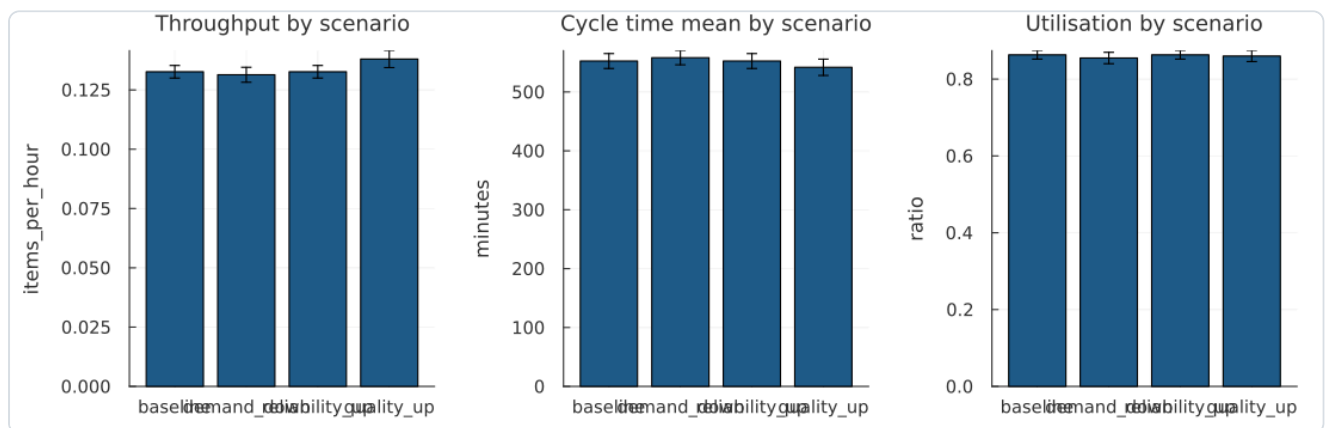
Value	Throughput mean	Wait mean	Utilisation mean	Cycle time mean
0.055	0.133	—	0.865	—
0.075	0.133	—	0.865	—
0.095	0.133	—	0.865	—
0.11	0.133	—	0.865	—
0.125	0.133	—	0.865	—

The best arrival_rate by cycle_time_mean is 0.055 (mean 552.0 ± 31.5).

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.133	0.131	-0.00126	0.00163	-0.953	indistinguishable
demand_down	cycle_time_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	wait_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	utilisation	0.863	0.855	-0.00824	0.00893	-0.955	indistinguishable
demand_down	completed	477.5	473.0	-4.5	5.844	-0.942	indistinguishable
reliability_up	throughput	0.133	0.133	0	0	0	indistinguishable
reliability_up	cycle_time_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	wait_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	utilisation	0.863	0.863	0	0	0	indistinguishable
reliability_up	completed	477.5	477.5	0	0	0	indistinguishable
quality_up	throughput	0.133	0.138	0.00535	0.00154	4.032	better
quality_up	cycle_time_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	wait_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	utilisation	0.863	0.86	-0.00318	0.00578	-0.369	indistinguishable
quality_up	completed	477.5	496.8	19.2	5.523	4.031	worse



Scenarios

Best by cycle_time_mean: **quality_up** (compared with baseline, common random numbers on every replication).

The models

Five models, one engine: a service pool, a transfer line, a job shop, an inventory position and a contact centre.

Models

Model	Entity	Completed	Throughput	Cycle time	Wait	Utilisation	Availability
mmc	customer	3048	0.847	1.912	0.582	0.563	1.0
transfer_line	part	711.9	0.198	89.7	89.7	0.832	0.974
machine_shop	job	477.5	0.133	552.5	552.5	0.863	1.0

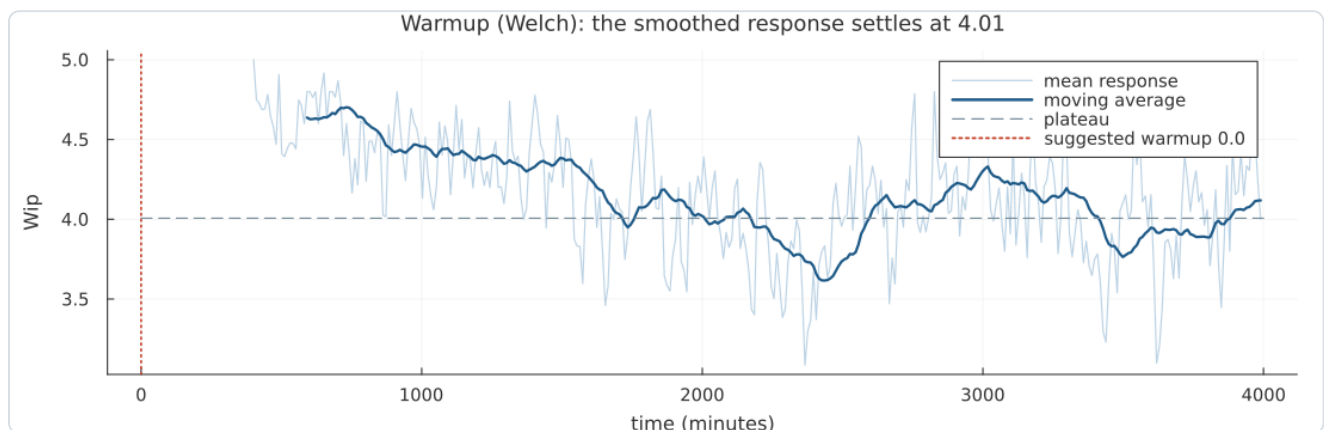
Model	Entity	Completed	Throughput	Cycle time	Wait	Utilisation	Availability
inventory	order	7178	1.994	—	—	—	—
call_center	call	3041	0.845	1.341	0.011	0.281	1.0

Experiments

Replications, a detected warmup, batch means for a single run, paired comparisons and a factorial design -- all with common random numbers.

How long the transient lasts

Welch's procedure on the wip series over 5 replications suggests discarding the first 0 time units (plateau 4.007 $\pm$ 0.088).



Warmup

One long run, honestly

Batch means

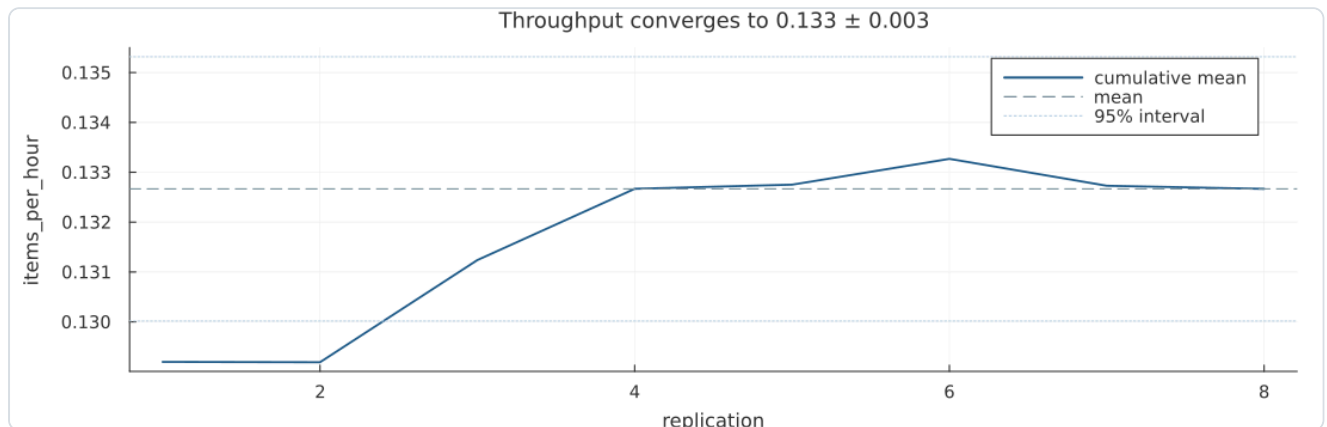
N	Mean	Half width	Lo	Hi	Batches	Lag1	Adequate
8	552.5	8.555	544.0	561.1	4	-0.543	no

The replications

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
wait_mean	552.5 none	12.7 none	minutes	2.294
wait_p95	1034 none	103.3 none	minutes	9.987
cycle_time_mean	552.5 none	12.7 none	minutes	2.294
service_mean	— none	— none	minutes	—
completed	477.5 none	9.543 none	count	1.998
throughput	0.133 none	0.00265 none	items_per_hour	1.998
wip_mean	4.155 none	0.061 none	count	1.476
tardiness_mean	449.0 none	11.5 none	minutes	2.564
scrapped	22.9 none	4.057 none	count	17.7
reworked	71.2 none	5.031 none	count	7.061

Metric	Mean	Half width	Unit	Rel hw pct
setup	2.0 none	0 none	count	0
utilisation	0.863 none	0.011 none	ratio	1.285
availability	1.0 none	0 none	ratio	0
max_queue_length	44.1 none	6.49 none	count	14.7
queue_length_mean	75.5 none	0.054 none	count	0.071
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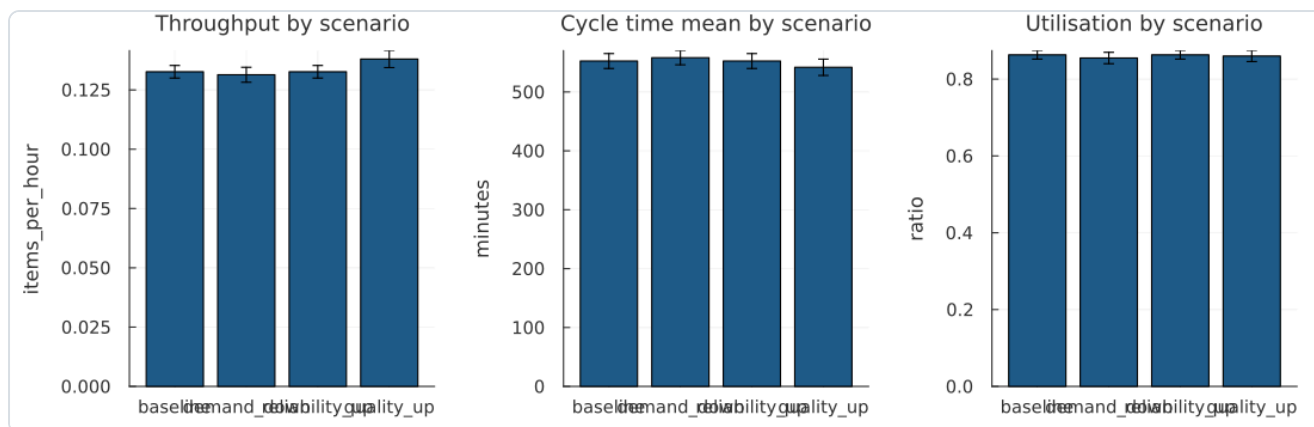


Convergence

Scenarios against the baseline

Scenarios

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demand_down	throughput	0.133	0.131	-0.00126	0.00163	-0.953	indistinguishable
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reliability_up	utilisation	0.863	0.863	0	0	0	indistinguishable
reliability_up	completed	477.5	477.5	0	0	0	indistinguishable
quality_up	throughput	0.133	0.138	0.00535	0.00154	4.032	better
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quality_up	wait_mean	552.5	541.9	-10.6	5.537	-1.926	better
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quality_up	completed	477.5	496.8	19.2	5.523	4.031	worse



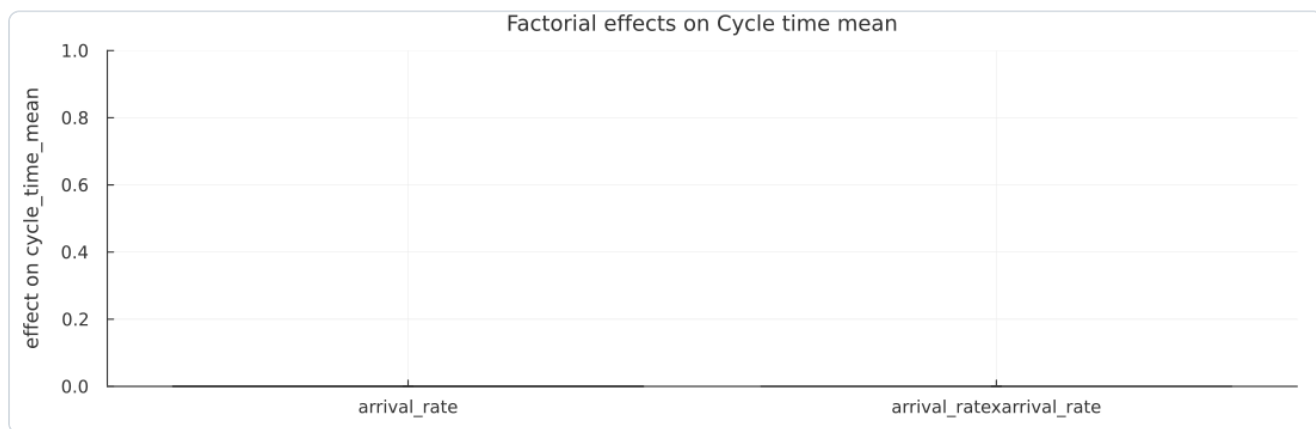
Scenarios

Best by cycle_time_mean: **quality_up** (compared with baseline, common random numbers on every replication).

Which knob matters?

Factorial

Term	Metric	Effect	Half width	Lo	Hi	Verdict
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_ratearrival_rate	cycle_time_mean	0	0	0	0	negligible



Factorial

Calibration

The model parameters come from measurements, and every fit is tested against the observations it claims to describe.

The observations

Plant history

Series	N	Mean	Sd	Cv	Min	P50	P95	Max	Lag1
interarrival	54127	1.197	1.218	1.017	7.96e-6	0.825	3.595	15.1	0.019
service	48780	1.328	0.627	0.472	0.171	1.201	2.519	8.086	0.0012
failure_interval	354	172.3	113.9	0.661	3.02	148.8	387.4	643.5	-0.038

Series	N	Mean	Sd	Cv	Min	P50	P95	Max	Lag1
repair	354	11.1	6.141	0.553	2.991	9.591	22.4	42.3	-0.014
demand_size	82	14.4	8.029	0.556	4.547	12.9	23.2	50.3	-0.139

Days 45.0

Base rate 0.8

Service mean 1.2

Failure mtbf 180.0

Drift 0.25

Maintenance day 45.0

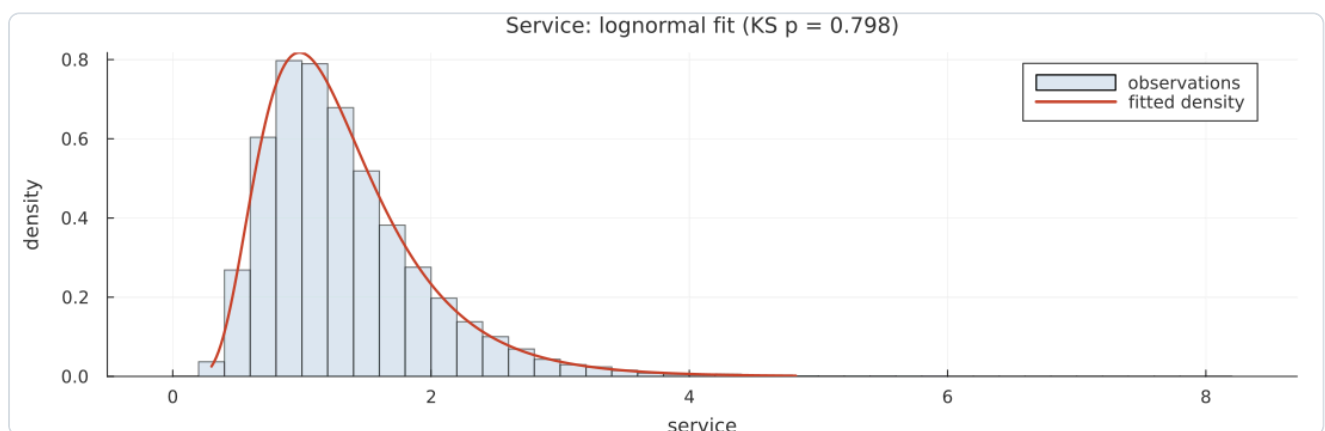
Service improvement 0.12

Seed 20260101

Maximum-likelihood fits

Fits

Series	Family	Mean	Sample mean	Cv	N	Ks stat	Ks p	Verdict
interarrival	gamma	1.197	1.197	1.005	54127	0.00293	0.741	validated
service	lognormal	1.329	1.328	0.473	48780	0.00292	0.798	validated
failure_interval	weibull	172.3	172.3	0.663	354	0.024	0.988	validated
repair	lognormal	11.1	11.1	0.529	354	0.044	0.488	validated
demand_size	lognormal	14.3	14.4	0.488	82	0.056	0.957	validated



Calibration

What the data implies for the model

Arrival rate 0.835

Service rate 0.753

Mtbf 172.3

Mttr 11.1

Availability 0.94

Demand mean 14.3

Offline and online

The same observations, refreshed periodically, with the decision they justify written down.

The feed

Url https://raw.githubusercontent.com/caxelrud/Julia_Discrete_Sim_2026/main/data/online_feed.json

Policy offline_first

Status cached

Source offline_cache

Fetched at 2026-09-24T09:45:47

Freshness unknown

Attempts 2

Error TypeError: in keyword argument connect_timeout, expected Int64, got a value of type Float64

Fallback local_file

What arrived

Feed observations

Series	Observations
interarrival	4000
service	4000
failure_interval	354
repair	354
demand_size	82

Offline against online

Reevaluation

Parameter	Offline	Online	Relative change	Beyond threshold
arrival_rate	0.835	0.796	-0.047	no
service_rate	0.753	0.754	0.00238	no
mtbf	172.3	172.3	0	no
mttr	11.1	11.1	0	no
demand_mean	14.3	14.3	0	no

Are the two samples the same process?

Two sample ks

Series	Ks stat	Ks p	N offline	N online	Same process
interarrival	0.019	0.125	54127	4000	yes
service	0.012	0.634	48780	4000	yes
failure_interval	0	1.0	354	354	yes
repair	0	1.0	354	354	yes
demand_size	0	1.0	82	82	yes

The reevaluation log

Log

At	Source	Status	Freshness	Age days	Observations	Worst change	Verdict	Reason
2026-09-24T04:45:49	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold

The model always answers from local data first; the feed only ever improves the picture, so nothing here fails when the network is down.

Artefacts

Every number, figure and PDF in this repository is regenerated by one command from one seed.

Data

Data

File	Directory
observations_interarrival.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_service.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_failure_interval.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_repair.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
observations_demand_size.csv	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
history.json	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
calibration.json	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
online_feed.json	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data
analysis.json	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\data

Figures

Figures

File	Directory
wip.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
wait.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
utilisation.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
throughput.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
gantt.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
scenarios.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
sweep.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
warmup.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
validation.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
factorial.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
calibration.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures
convergence.png	C:\Users\inter\OneDrive\myWork\Research2026\Julia_Discrete_Sim_2026\reports\figures

Reproduce everything with one command and one seed: `julia --project=. scripts/run_study.jl --seed=20260101`.

Generated by DiscreteSim.jl from seed 20260101; every number is reproducible.